

Clean and secure power supply: A system dynamics based appraisal

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ARTICLE INFO

Keywords:

Security of supply
Renewables
Electricity markets
System dynamics

ABSTRACT

Across the world, the electricity industry is changing with the advent of renewable energy sources such as solar and wind power. As these are exposed to intermittency, seasonality, and global-climate-variation, policy-makers' concerns are now shifting towards security of supply. In fact, the industry confronts three conflicting goals: security of supply, competitive prices to customers, and environmental protection. In this context, and given the multiple uncertainties of technology transformation, this paper uses modelling-based scenario analysis to investigate different extreme and plausible futures. Simulation is used to analyse policies aimed at increasing the penetration of renewables and to explore how these energy sources may affect system reliability. This paper explores the effect of incorporating renewables in Colombia, where a large hydroelectric component has led to insufficient electricity being available during droughts and to high electricity price volatility. Though not intuitive at first glance, this paper shows that renewables may contribute to: i) increased security of supply through complementarity, e.g., it does not rain when the sun shines; ii) reduced price volatility in the medium-term, and iii) increased industry sustainability.

1. Introduction

Prior to the 1990s, governments favoured monopolies in power supply industries for both the strategic value of electricity and the economy of scale argument. However, as monopolies led to inefficiencies along the supply chain, the industry was liberalized and opened to competition in both the generation and supply of electricity. These changes motivated, on the one hand, improvements in service and quality (de Vries and Heijnen, 2008; Newbery, 2002; Stoft, 2002), but on the other hand, largely induced industry concentration and, consequently, failed to transfer electricity cost reductions to customers in an appropriate manner (François et al., 2016; Joskow, 2008a). Having not attained the goals of these reforms yet, current worldwide climate alerts have prompted government agreements on reducing greenhouse gases (GHGs), thereby promoting novel changes in the electricity industry, the largest GHGs emitter (IEA, 2017; IPCC, 2014; PBL, 2016). Broadly, policy makers are currently seeking to achieve three main objectives: (i) efficient markets, based on low carbon sources e.g. wind and solar, (ii) security of supply, and (iii) affordable prices to consumers (World energy council, 2013).

As a consequence of these changes, renewables, such as solar and wind power, are spreading, with implications for security of supply because of their intrinsic intermittency (Battaglini et al., 2009). This

poses questions regarding the extent of the dissemination of renewables, their effect on security of supply and their economic impact. However, the extent to which these questions have been explored varies from country to country.

Our research focuses on the Colombian electricity sector, which is characterised by a large hydro-based technology component that is exposed to droughts, *El Niño*-Southern Oscillation (ENSO), and floods, *la Niña*, increasing the volatility in both electricity supply and price (Dyner et al., 2007; XM, 2018). Though Colombia's prospects in hydro, wind and solar power are among the most important in Latin America (Caspary, 2009), Colombia has not yet adequately developed renewable technologies, and the implementation of a low-carbon strategy is still in the initial stages (Akif and Aslan, 2017; Edsands, 2017; Ministerio de Minas y Energía, 2017). As has been proven, there are significant complementarity between hydropower and other renewable energies, including solar and wind (Cantão et al., 2017; de Jong et al., 2017; Denault et al., 2009; Henao et al., 2019; Pianezzola et al., 2017; Schmidt et al., 2016, 2015). This paper thus assesses how the integration of these energy sources may contribute to security of supply, environmental sustainability and affordable prices for consumers.

Given the number of variables involved, their entangled relationships and the non-linear feed-back effects among them, this paper examines, for the mid-term, the broad impacts of the penetration of

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renewables as well as their complementarity with hydropower. In this complex context, a model-based approach is proposed for assessing, in terms of the environmental sustainability and price affordability, the effects of a policy promoting renewables. There are countries with large hydropower systems or with a high share of renewables such as Brazil, Costa Rica, New Zealand, Denmark, Iceland, Norway, and Uruguay (Prado et al., 2016; De Oliveira-De-Jesus, 2019; IEA, 2018; Statistics Norway, 2018). In this paper, Colombia was chosen for the case study due to it being a country with a very large hydropower system in place, and one that is also gifted with abundant renewable energy resources (Cardenas et al., 2016; Pérez and Osorio, 2002; UPME, 2005).

The main objective of this research is to undertake a novel model-based-assessment approach to examine the effect of the diffusion of renewables and their complementarity on the overall energy sector. Furthermore, this paper seeks to provide a better understanding of the effect of the diffusion of renewable energies on the following indicators: CO₂ emissions, system reserve margin (a proxy for security of supply), and electricity price. With the support of a simulation model, this paper contributes to the evaluation of the implications of the diffusion of renewables in Colombia, considering renewable energy sources' complementarity, security of supply and CO₂ emissions. The paper assesses whether there would be beneficial effects for Colombia in utilising renewables, and contributes to the enrichment of the understanding of the future of renewables not only in Colombia, but also around the world.

The paper is organised as follows. The next section provides an overview of the literature on the penetration of renewables in electricity markets, with emphasis on security of supply, the focus of our study. Next, we discuss the Colombian electricity industry. The fourth section describes a System Dynamics (SD) model that has been built to study the dynamics of both the actual and the potential market, and that includes representations of policies to increase renewable energy supplies in Colombia. The penultimate section discusses simulation results and policy analysis, and conclusions are drawn in the last section.

2. Assessing the effect of the penetration of renewables in electricity markets

Renewable energies are sources that are continually replenished by nature. While hydropower is a conventional renewable technology that has been used widely for over a century (Ellabban et al., 2014), wind and solar are non-conventional and although intermittent they have 'only' slight social and environmental impacts (Khan, 2006).

The penetration of renewable energy sources and their effect on security of supply have been explored in a variety of contexts, including geographic location, degree of development, policy and regulation. In addition, renewable energies benefit from their complementarity, which help guarantee both electricity supply throughout the day and also price affordability. This section examines these issues, based on specialised literature.

Renewable energy industries have boomed since 2008, with wind and solar displaying the highest growth rates. Installed wind energy worldwide grew swiftly from 6.1 GW in 1996 to 539 GW in 2017, an annual-average growth rate of 20% (Joshi et al., 2009; REN21, 2018; Zhou et al., 2016). Solar energy has been growing at rates up to 33% annually in recent years due to solar photovoltaic (PV) cost-reduction and world concern about climate change, the current installed solar PV capacity is 402 GW (OECD and IEA, 2014; REN21, 2018) – these diffusions have been taking place despite the intrinsic intermittency of the technologies. Owing to the fact that new generation is therefore increasingly subject to intermittency, capacity mechanisms are essential for security of supply in the long term (Cantão et al., 2017). Although renewable energies represent a challenge, countries are adopting these technologies and some are even committed to achieve 100% renewable, e.g. Denmark, Germany, the UK and New Zealand have announced a

long-term target of 100% independence from fossil fuels and nuclear power (Lund and Mathiesen, 2009; Mason et al., 2013; Palzer and Henning, 2014; Procter et al., 2018; Walmsley et al., 2017).

Given that the intermittency of generation increases with the addition of renewables – at short notice wind speed falls or sun fades – scholars are investigating alternative incentives for capacity investment to secure electricity supply (Castro-Rodriguez et al., 2009; Cramton et al., 2006; García-gusano et al., 2017; Gouveia et al., 2014; Hasani-Marzooni and Hosseini, 2013; Joskow, 2008b; Moreno and Martínez-Val, 2011; Rodilla and Batlle, 2012). These papers conclude that though, on the one hand, renewables pose challenges to electricity markets, on the other, it is feasible to achieve security of supply.

Renewable energy sources increase the portfolio of generation technologies but reliability concerns become more severe for power system planners. Researchers continue to investigate alternatives for the integration, optimisation, and complementarity of various renewable energy sources (de Jong et al., 2016; Elliston et al., 2014, 2013; Henao et al., 2019; McPherson et al., 2017; Oree et al., 2017). These papers indicate that a richer mix of energy sources is fundamental for reducing power system vulnerabilities and increasing energy independency from fossil fuels.

Security of supply is not sought solely via back-up thermal or additional renewable capacity, it may be achieved, at least partly, by making efficient use of the complementarity between renewable energies (Beluco et al., 2012; Bilil et al., 2014; De Andrade Meireles et al., 2013; De Jong et al., 2013; Gárdos et al., 2010; Kruangpradit and Tayati, 1996; Monforti-ferrario et al., 2016; Schmidt et al., 2016; Souza et al., 2017).

The literature discusses the penetration of renewables and their effects on security of supply; it also discusses the necessity for, and influences of, capacity mechanisms in the presence of a high share of renewables. In addition, some research focuses on the optimized use of renewable energy technologies. However, this paper found a gap in the literature regarding a model-based policy assessment on how the complementarity between solar and wind technologies with hydropower may contribute to energy efficiency and electricity price reduction, as a result of the negligible or almost zero variable costs (operation and fuel costs) of renewables (Helm, 2017), and energy independence from fossil fuels. This paper provides a novel approach that fills the gap and contributes to the improvement of knowledge and understanding with respect to renewables within large hydropower systems, particularly in Colombia.

This paper chooses the SD approach over optimisation, given the uncertainties and nonlinearities involved and its capability of modelling highly dynamic power markets, which are characterised by investment cycles that involve lags, nonlinearities and feedbacks, thereby making it possible to understand medium- and long-term policies and their effects on the system. Additionally, the SD methodology offers advantages over other alternatives as it helps to: (i) visualise the interactions and causal relationships between different variables, (ii) understand the impact of delays on the evolution of the system, and (iii) assess the response of the system under different scenarios (Dyner and Larsen, 2001; Osorio and van Ackere, 2016; Perez and Osorio, 2002; Ponzo et al., 2011; Sterman, 2000).

While some recent research has used SD to assess the effect of renewable energies on security of supply and energy-emissions, those papers did not consider how the combinations of hydroelectricity, solar and wind affect the system in the medium-to long-term (Cardenas et al., 2016; Franco et al., 2015; Osorio and van Ackere, 2016). Considering the complementarity between solar, wind and hydropower, we use SD to investigate the integrated effect of alternative policy instruments on electricity price and on the system reserve margin in Colombia, an energy-resource-rich country that, with the exception of hydro, has not yet significantly incorporated clean energies into its power matrix.

3. Challenges for the diffusion of renewable energies in Colombia

To better understand the focus of this research, this section briefly discusses the main challenges for the diffusion of renewable energies in Colombia. Hydropower currently accounts for 66% of the total installed capacity, while run-of-the-river power plants account for 3.7%, and thermal electricity, comprising Combined Cycle Gas Turbine (CCGT) and coal plants, accounts for 30%. It is important to point out that the power industry is recurrently affected by severe weather variations, which reduce water inflows by up to 50%, making electricity prices volatile (Dyner and Larsen, 2001; XM, 2017).

Colombia implemented a capacity mechanism – a reliability charge – to increase capacity investment with the purpose of reducing the risk of blackouts during dry seasons (CREG, 2006). The reliability charge seeks to dampen cycles of over- and under-investment and avoid blackouts by stabilising generators' income and reducing price volatility, particularly in this hydro-based system (Cramton and Stoft, 2007; Dyner et al., 2007).

The capacity mechanism has unintendedly incentivized hydropower projects, which are mostly dependent on unpredictable weather conditions (Cramton et al., 2006; XM, 2018). Questions arise regarding the effectiveness of the reliability charge as a guarantee of system adequacy and, also, with respect to the impact of renewable energies in the Colombian electricity market.

Though Colombia has been heralded as one of the best markets in terms of electricity supply security and environmental sustainability (World energy council, 2013) it has repeatedly suffered from droughts with great threats to security of supply, in recent decades the worst of which occurred in 2015–2016 (Super Intendencia de servicios publicos, 2016). This has been the case because of the large dependency on hydropower, an energy source that is affected by unpredictable reductions of water inflows into reservoirs (Dyner et al., 2007) that result from the macroclimate phenomenon called El Niño-Southern Oscillation (ENSO). Consequently, prices increase considerably because hydropower generation is greatly diminished and demand has to be met using limited thermoelectric generation (Battile et al., 2010).

In Colombia, four factors constitute major risks for investments in new generation projects: a) electricity price volatility; b) the ability of hydroelectricity to meet all the electricity demand in the country, under average weather conditions, and discounting grid imperfections; c) climate seasonality, and d) the recurrence of El Niño. This evidences the problems of securing supply and the grounds for implementing a capacity mechanism that seeks to promote investment in new generation capacity through the stabilization of generator income (Rodilla et al., 2011). Fig. 1 shows electricity prices during a period that includes extreme droughts, such as those experienced during June 2015 and April 2016; prices reached as high as USD 1 per kWh for some days in

2015, and reached averages as high as USD/MWh 0.25 over several months in 2016. These exceed the maximum price paid by the electricity price (around USD/MWh 0.15) – scarcity price (price cap) – which means that several utilities lose vast amounts of money, threatening system sustainability.

Fig. 1 shows spot prices much higher than the scarcity price as, during the long dry season, water in reservoirs was depleted and fuel oil plants were needed to guarantee electricity supply, increasing the prices of the marginal plant by displacing those gas-fuelled plants (as no sufficient gas was available for their operation) from the merit-order (theory where the power offer made by each technology is ranked from the lowest to the highest price – forming the supply curve – and the price is determined by the cost of the last technology in meeting electricity demand (XM, 2018)). These facts are the basis of this research.

3.1. Renewable energy potentials of Colombia

The current renewable-energy installed capacity in Colombia is 491.5 MW, of which only 19.5 MW comes from wind energy, 463 MW from small hydropower plants, and 10 MW from photovoltaic solar systems (XM, 2018). Colombia has a great potential for hydroelectricity; the estimated small hydropower potential is 25,000 MW. The potential of wind power in the La Guajira region is 18,000 MW. In addition, in Guajira region is among the best wind qualities in South America (10 m/s average) (Caspary, 2009; Vergara et al., 2010). In Colombia, the average solar irradiation is 4.5 kWh/m²/day; in La Guajira region the solar irradiation is around 6.5 kWh/m²/day, greater than the average solar irradiation of China where it is 4.5 kWh/m²/day (Che et al., 2005; Corpoema, 2010; EPIA, 2014; IDEAM, 2016; Martino et al., 2013). However, China had the highest cumulative installed-capacity of solar PV in the world in 2017 (REN21, 2018).

As indicated, despite these potentials that are so noteworthy, not only in the context of Latin American but also worldwide (Caspary, 2009), Colombia has not yet developed them sufficiently. Contrary to public beliefs, this country is not taking advantage of the complementarity among renewables that may help to reduce prices and increase security of supply (Cardenas et al., 2016).

Fig. 2 shows the complementarity between water inflows and wind speed in Colombia, and Fig. 3 shows the complementarity between solar and energy inflows. In this context, this research explores the great opportunities that these energy sources offer for energy efficiency, security of supply and affordable electricity prices, mostly by using solar and wind power during dry seasons, and by generating electricity with hydro during the rainy seasons, while also keeping some potential energy-generation water stored in reservoirs for dry periods.

Fig. 3 shows the complementarity of the average solar availability (hr/day) and water inflows (m³ per month) – for most of the time,

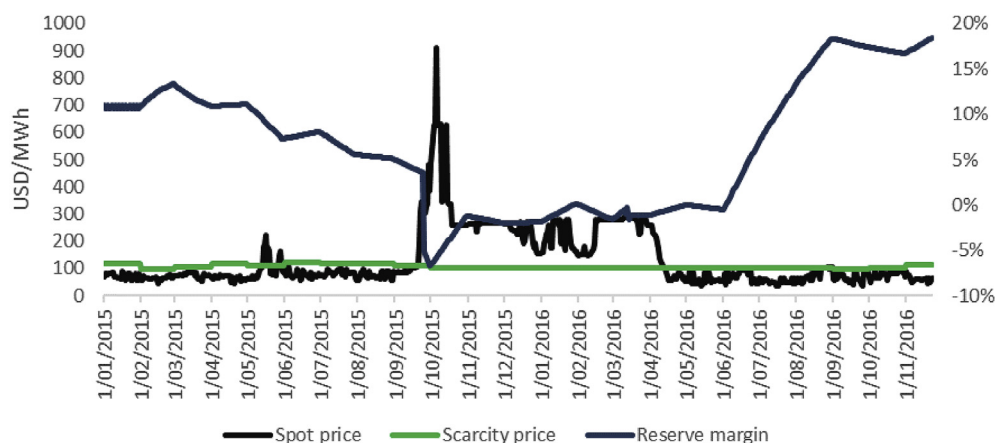


Fig. 1. Spot electricity price in Colombia, taken and adapted from (XM, 2018).

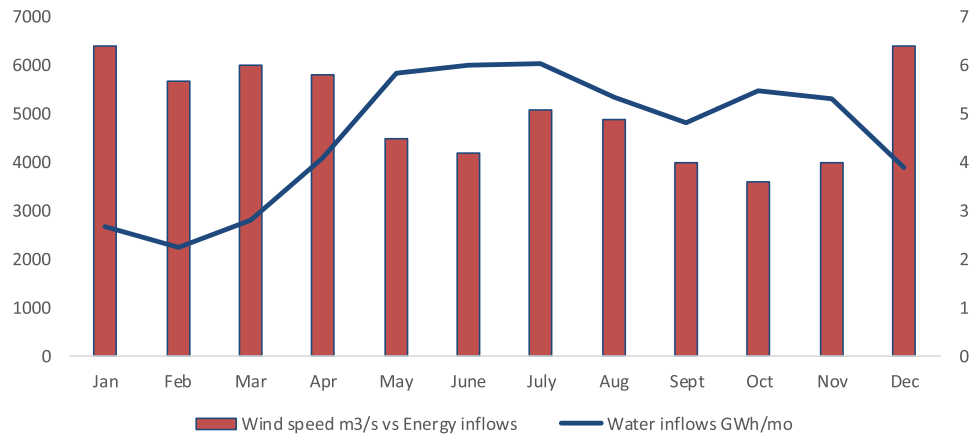


Fig. 2. Wind and hydropower complementarity – data taken from (IDEAM, 2016).

when water inflows decrease then solar increases, particularly between January–May and October–December. A high level of renewable power capacity along with renewable complementarity leads to electricity price reductions, because when water reservoirs are high, water can be saved and used during the following seasons, when solar and wind power are not available (Zapata et al., 2018). Due to not being the focus of the paper, neither hourly complementarity nor broad detailed complementarity analyses are reported here (Henao et al., 2019).

For dispatch and reliability purposes some models require a very tight time scale (Blakers et al., 2017; Bogdanov and Breyer, 2016; Elliston et al., 2014; Esteban and Portugal-Pereira, 2014; Gulagi et al., 2017a; Lund and Mathiesen, 2009; Palzer and Henning, 2014; Sadiqa et al., 2018; Heard et al., 2017). However, as the focus of the paper is medium-to long-term, the model here has a monthly resolution. Furthermore, the model does not need to be more detailed as storage (here in the form of hydro reservoirs) provides a solution to the intermittent nature of solar PV and wind.

Blakers et al. (2017) analyse the Australian case considering the support of hydroelectricity and biomass to guarantee system reliability. Gulagi et al. (2017b), François et al. (2016), Lund et al. (2015), and Scorah et al. (2012) indicate that system reliability may be attained with renewables in India and elsewhere.

Against this background, our study examines, through simulation, whether the complementarity among renewables could promote energy efficiency, which may lead to economic efficiency – and could help address the question: Might the Sun and wind power, together with water inflows into hydroelectricity reservoirs, help improve system reliability? A broad view on renewables may help understand how complementarity could operate in the power supply industry. When taking a mid-to long-term perspective regarding these issues, modelling

is of significant value as there are cumulative behavioural effects that need to be taken into account under different scenarios. The next section describes the SD model that was developed by the authors of this study for assessing the mid-to long-term effect of alternative policies that promote renewable energies to assist with security of supply in Colombia.

4. Modelling for policy assessment

This section proposes an SD model for the Colombian power sector that incorporates some of the key variables of the system, such as the evolution of power capacity by technology, CO₂ emissions, system reserve margin, and generation price. These variables are required as the aim of the paper is to assess whether the electricity market is capable of meeting the following goals: (i) clean generation, replacing thermal generation by wind and solar, (ii) security of supply, and (iii) affordable prices to consumers.

4.1. Dynamic hypothesis

Fig. 4 shows the dynamic hypothesis of this research. When markets operate, increases in both demand and supply contribute, in opposite ways, to the reserve margin of the system (i.e. the difference between supply and demand), closing both of the control cycles that contribute to oscillations of the reserve margin (Dyner, 2000; Larsen and Bunn, 1999). Furthermore, as the reserve margin increases so electricity price decreases, incentivising investment in new capacity generation, in both conventional power capacity as well as renewable power capacity. The hypothesis comprises three cycles. Balancing loop B1 establishes the mid-term demand elasticity to price. Balancing loops B2 and B3

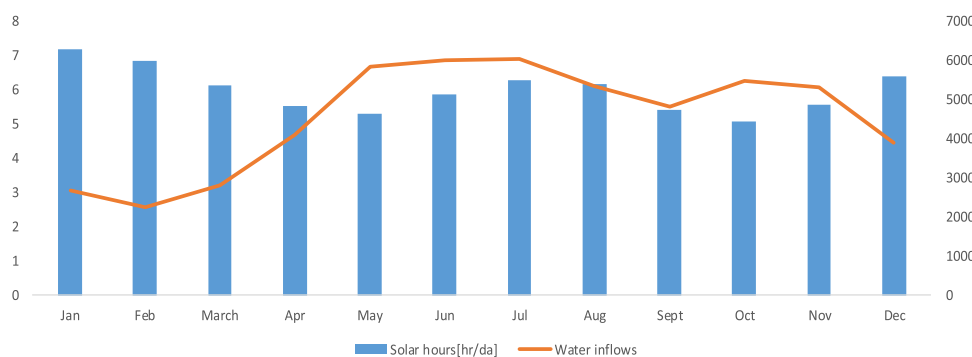


Fig. 3. Solar and hydropower complementarity – data taken from (IDEAM, 2016).

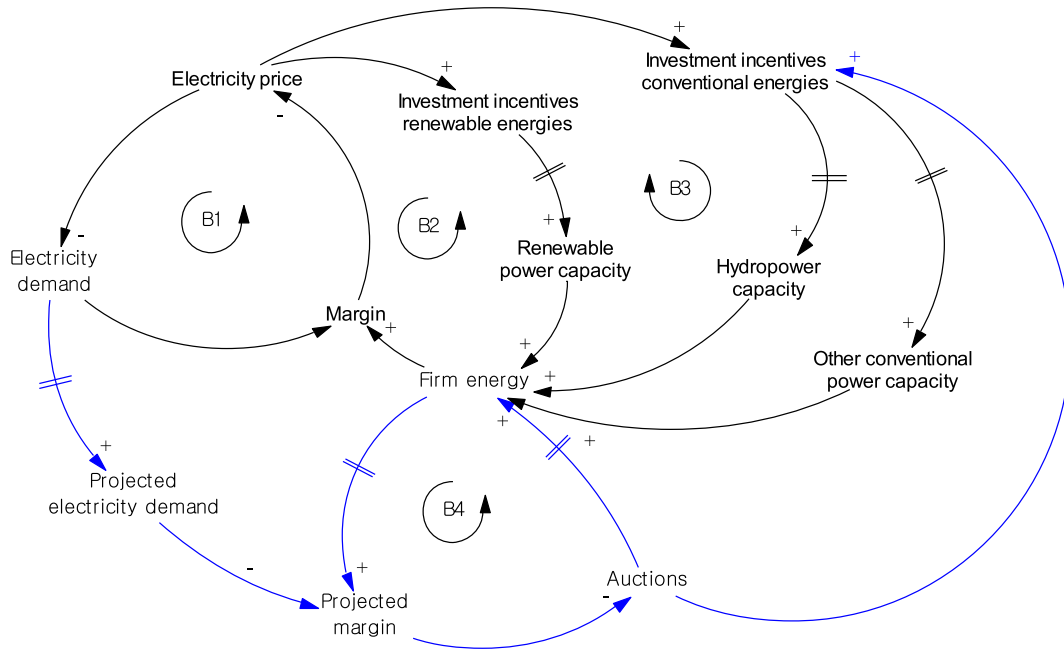


Fig. 4. Dynamic hypothesis for the Colombian electricity market with renewable energies.

establish how electricity generation affects margin and electricity price, which creates incentives for new power capacity. In addition, the complementarity between renewable energies and hydropower induces energy firmness, increasing energy efficiency and reducing electricity prices. However, given the intermittent or seasonal characteristics of renewable power capacity, this paper investigates whether policy is required to guarantee security of supply (Franco et al., 2015). In addition, capacity mechanism is modelled in balancing loop B4, if both investments in renewable energies and the projected reserve margin drop, reservoir-based hydro- and thermo-electricity will provide firmness, with the support of auctions, by increasing the reserve margin.

4.2. Stocks and flows diagram

To analyse the Colombian market dynamics, this section next presents the main stocks and flows diagram and some of the key corresponding equations of the simulation model. The model is composed of three modules: supply, price formation and policy.

The electricity supply component includes available renewable technologies (solar and wind), the installed capacity in place, and their main resulting variable: generation price. Fig. 5 presents the model structure of the supply side. The balance between demand and supply determines the formation of price in the model. Each of the technologies offers quantity and price. These bids are organised according to merit order, from the lowest to the highest price, thus establishing the corresponding supply curve (Corpoema, 2010; Dyer et al., 2013).

The main equations are described next. Eq. (1) establishes that capacity under construction (CUC) depends on the new capacity to construct (NCC) and construction completion (CC); Eq. (2) indicates that installed capacity (IC) depends on construction completion (CC) and capacity depreciation (CD); an Eq. (3) shows that (NCC) depends on new capacity investment, which is led by Investment by profitability (IBP) and Capacity markets (CMI).

$$CUC(t) = CUC(0) + \int CC(t) + NCC(t) - CT(t)dt \quad (1)$$

$$IC(t) = IC(0) + \int CC(t) - CD(t)dt \quad (2)$$

$$NCC = IBP(t) + CMI(t) \quad (3)$$

4.3. Main assumptions and data used

The simulation model is based on the following assumptions:

- For simplicity, power transmission has no restrictions.
- For economic and environmental impact analysis of the different technologies involved, this research considered levelised cost of electricity by technology, capital investment and CO2 emission during usage; with these according to environmental regulation in Colombia (MINISTERIO DE AMBIENTE Y DESARROLLO SOSTENIBLE, 2017; UPME, 2016).
- The relevant time horizon for simulation is 2017–2039, which is sufficiently long to assess the dynamics of the market and to analyse the effects on security of supply and affordability. The simulation step is monthly. The drivers of the model are the investment decision processes relating to power generation and they depend on: a) tariff formation, b) renewable complementarity and c) choice of generation technology.
- Initial conditions have been taken from official entities of the power sector, including *Unidad de planeación minero energética* (UPME) and *Expertos en mercados* (XM).
- Feed-in-tariff (FIT) is considered endogenously determined for wind and solar systems (non-conventional technologies) (Newbery, 2012).
- Because of model specifications, this paper only considers monthly (seasonal) and ENSO (yearly) complementarity between solar, wind and hydropower (Henao et al., 2019).
- Electricity demand increases according to estimates by UPME (2016).
- The paper assumes that the national PV learning-curves have the same behaviour as the international ones.
- No strategic behaviour is modelled. The maximum electricity price (scarcity price) that can be achieved when the market is under extreme scarcity conditions is the generation cost of the most expensive technology (oil-based technology), and scarcity price is attained when the system reserve margin is close or less than zero.
- The capacity market considers gas power, coal power, oil and hydropower plants (UPME, 2016).

Table 1 synthesises other assumptions and input data of the model.

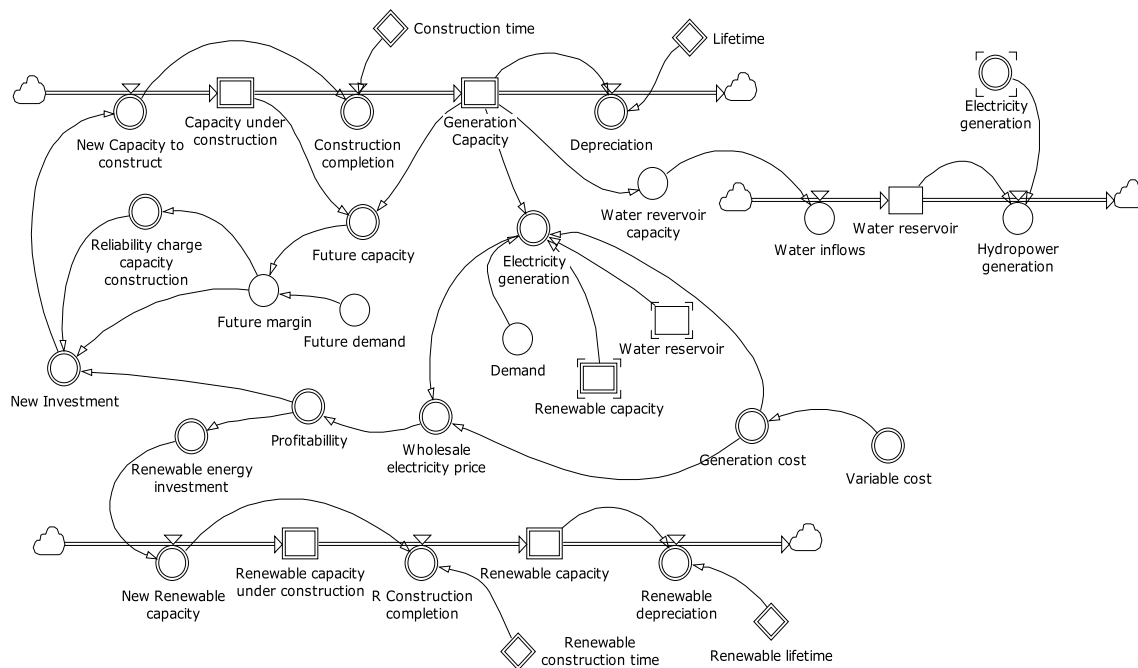


Fig. 5. Stocks and flows diagram of supply component.

Table 1

Main assumptions and input data in the model.

Capacity expansión						
New power capacity [MW]	Natural gas	Coal	Hydro	Fuel oil	Solar	Wind
	88	0	2400	354	0	0
New power capacity to construct depends on the profitability of each technology and the National Expansion plans. The reference values are estimated and based on the "Reference Expansion Plan Generation - Transmission 2014–2028" (UPME, 2014)						
Investment cost						
Investment cost [USD/KW]	Natural gas	Coal	Hydro	Fuel oil	Solar	Wind
	637	1387	1691	550	2243	1630
The reference values are estimated based on (USD/MWh) (eia, 2017; International Energy Agency, 2018)						
Emission factor						
Emission factor [Ton CO ₂ /MWh]	Natural gas	Coal	Hydro	Fuel oil	Solar	Wind
	0.41	0.9	0	0.7	0	0
The reference values are estimated based on (USD/MWh) (eia, 2017; International Energy Agency, 2018)						
Fossil fuels' costs						
Fossil fuels' costs [USD/MWh]	Natural gas	Coal	Hydro	Fuel oil	Solar	Wind
	11	14	0	100	0	0
The reference values are estimated and based on the "Reference Expansion Plan Generation - Transmission 2017–2035" (UPME, 2017)						
Other assumptions						
The estimated value of the feed-in-tariff is 50 USD/MWh for solar technology and 20 USD/MWh for wind technology.						
The estimated value of electricity demand growth is 2.9 per cent (UPME, 2016)						

4.4. Model validation

The model developed for policy analysis was extensively validated. This research uses several tests for consistency and conceptual validation purposes (Barlas, 1996; Sterman, 2000). Fig. 6 shows actual wholesale electricity price data versus simulated data – a model behavioural test. It can be seen that both sets of prices have the same trend, with a correlation coefficient $r = 0.9$; nonetheless, the model does not account for strategic behaviours or changes in operational conditions (as happened when CCGTs opted to switch from gas to diesel as there were not sufficient quantities of natural gas available in the market). This data is for the period 2015–2016 when the Colombian electricity market experienced one of the longest and driest seasons in 30 years.

Other validation tests which were conducted, but not reported due to space constraints, included: empirical structure and parameters, boundary adequacy, dimensional consistency and extreme conditions. Structurally oriented behavioural tests and sensitivity analysis were also conducted. More extensive validation of earlier versions of the model has been reported elsewhere (Zapata et al., 2018), proving how useful the approach may be for assessing *what if* policy scenarios.

The next section turns to policy analysis and discusses simulation results under different scenarios for the Colombian electricity market.

5. Policy analysis

To study the effect of the penetration of renewables, taking into account the potential complementarity among them, this paper studies four scenarios (Fig. 7) that result from considering two orthogonal policies, namely the promotion of renewables and the capacity mechanism policy. Each scenario analyses total installed capacity, renewable energies' share, CO₂ emissions, electricity price, electricity price volatility and system reserve margin.

First, the business-as-usual (BAU) scenario considers some elements of a free market (non-incentive schemes for renewable technologies such as wind, solar and small hydropower) together with a capacity mechanism (CM) to promote investment in new power capacity; the capacity mechanism predominantly incentivizes hydropower investment. Second, the Free Market (FM) scenario considers neither a capacity mechanism to support system expansion nor new transmission



Fig. 6. Wholesale electricity real price data versus simulated data.

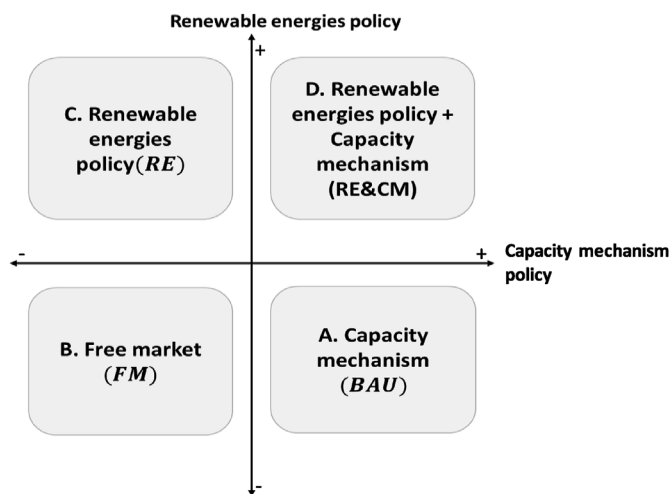


Fig. 7. Scenarios for analysing the Colombian electricity market.

lines for the promotion of renewable energies, the electricity price is the only driver of capacity investment (de Vries and Heijnen, 2008). The third, Renewable energies (RE) scenario, incorporates policy that

promotes renewables, a feed-in-tariff support scheme is used; however, a capacity mechanism is not used, i.e., the electricity price is the only signal to invest in conventional technologies. In general, the absence of a capacity mechanism may lead to low investment thereby increasing the risk for security of supply (De Vries, 2007). The fourth, Renewable energies plus capacity mechanism (RE&CM) scenario promotes renewables (using a feed-in-tariff scheme) with the support of backup power capacity to seek security of supply (Batlle and Pérez-Arriaga, 2008; Newbery, 2015). In this scenario, a capacity mechanism is taken into account to increase investment in conventional technologies. Besides these four scenarios, this paper also considers a fifth scenario for assessing the likely efficiency gains resulting from the complementarity between renewable sources. Feed-in-tariff support and net metering schemes are used to increase investment in non-conventional technologies; a capacity mechanism is applied for hydropower and small hydropower to take advantage of the complementarity between technologies.

Medium-to long-term simulation results for each scenario are discussed separately, taking into account the dynamics of the most relevant system indicators: security of supply (system reserve margin), environmental quality (carbon emissions) and, affordability (electricity price or unit electricity cost). These indicators have been chosen for the purpose of gauging policy costs and benefits.

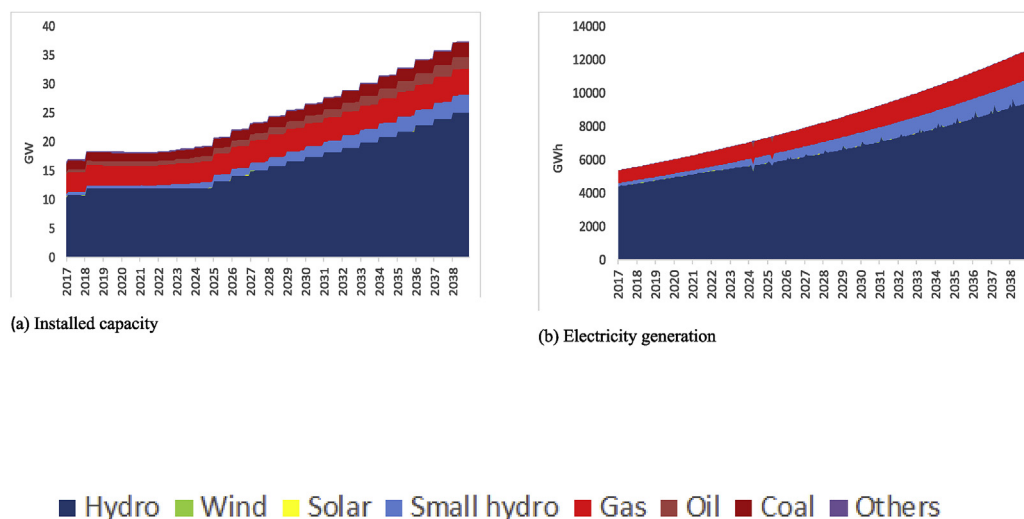


Fig. 8. (a) Installed capacity, with percentage of market share by technology to 2040, and (b) Electricity generation, in the BAU scenario.

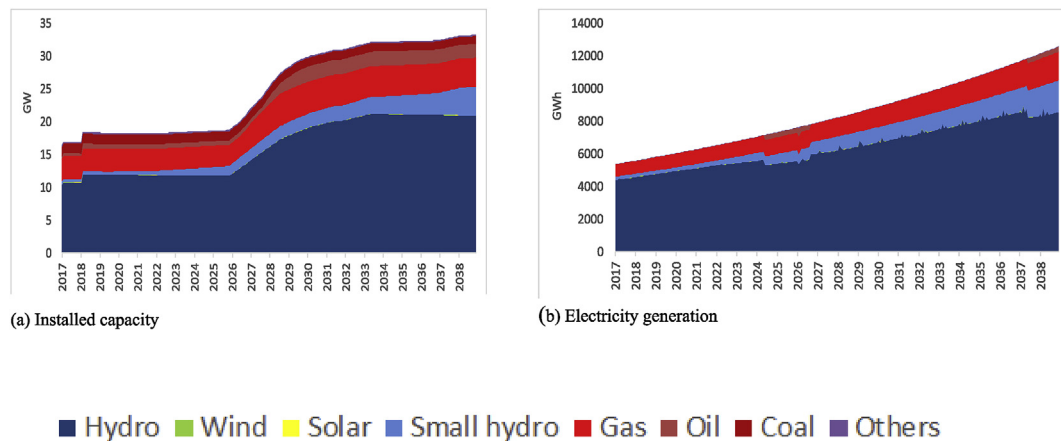


Fig. 9. (a) Installed capacity, with percentage of market share by technology to 2040 and (b) Electricity generation, in the FM scenario.

5.1. Business as usual (BAU)

Fig. 8 illustrates the simulated installed capacity (8a) and the simulated evolution of the monthly electricity generation (8b) for each of the modelled technologies. Fig. 8b shows that installed capacity grows continuously, following demand, largely supported by the capacity mechanism (reliability charge).

5.2. Free market scenario (FM)

This is similar to the previous scenario, except for the abolition of the capacity mechanism that induced generation capacity investment in Colombia. Fig. 9 illustrates the simulated installed capacity (9a) and the simulated evolution of monthly electricity generation (9b), for each of the modelled technologies.

Fig. 9a and b show that electricity generation increases while installed capacity remains unchanged between 2018 and 2026, as the system reserve margin was very high at the simulation outset. Fig. 9a shows small hydro technology increases due to its lower cost compared with other technologies.

5.3. Renewable energies scenario (RE)

This scenario analyses the inception of solar PV generation in the residential sector and a feed-in-tariff policy to increase the exploitation of renewable energy sources. As may be appreciated in Fig. 10a, solar, wind and hydro have important complementarity, but this is not

enough to guarantee security of supply, as during the simulation the system reserve margin is negative, which results in blackouts. Fig. 10 shows the simulated installed capacity (10a) and simulated evolution of monthly electricity generation (10b) for each of the modelled technologies, in which solar PV, wind and small hydro technologies grow dramatically; therefore, thermal generation drops.

5.4. Renewable energies scenario + capacity mechanism (RE&CM)

Fig. 11 shows the simulated evolution of the installed capacity (11a) and monthly electricity generation (11b) by modelled technology. Note that, in particular, the technologies of solar PV, wind and small hydro grow swiftly. Fig. 11b shows that thermal electricity and hydroelectricity grow during most periods of the simulation.

5.5. More Favourable Renewables Component scenario

Scenario D found that renewables, with some help from gas-fired plants, contribute greatly to the reduction of pool price and CO₂ emissions (Fig. 13c), while preserving an appropriate system reserve margin. Further, as previously discussed in the RE scenario, although the renewables policy does not avoid blackouts – which are bad – they are kept down to low levels. In this context, we investigate here whether a more favourable policy towards renewables, though not an obvious course of action, might nevertheless contribute to security of supply. Specifically, this section investigates whether the system reserve margin might improve, and also what the effect on electricity

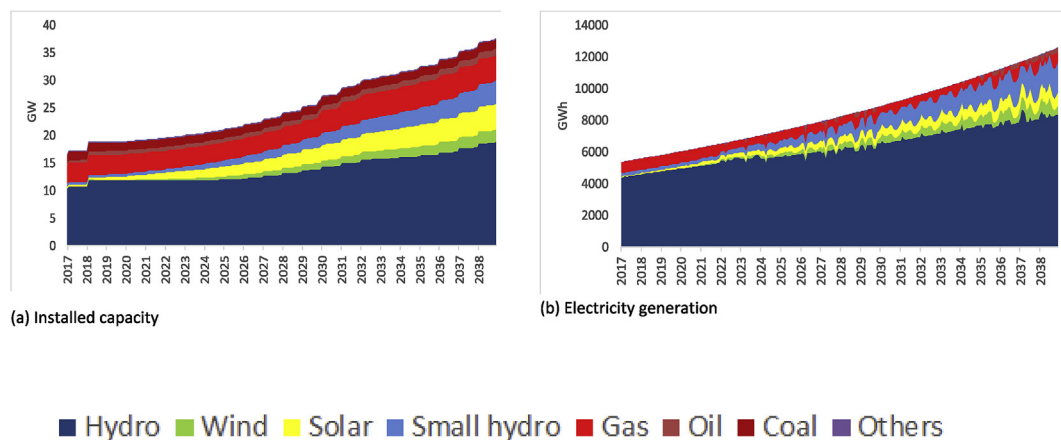


Fig. 10. (a) Installed capacity, with percentage of market share by technology to 2040, and (b) Electricity generation, in the RE scenario.

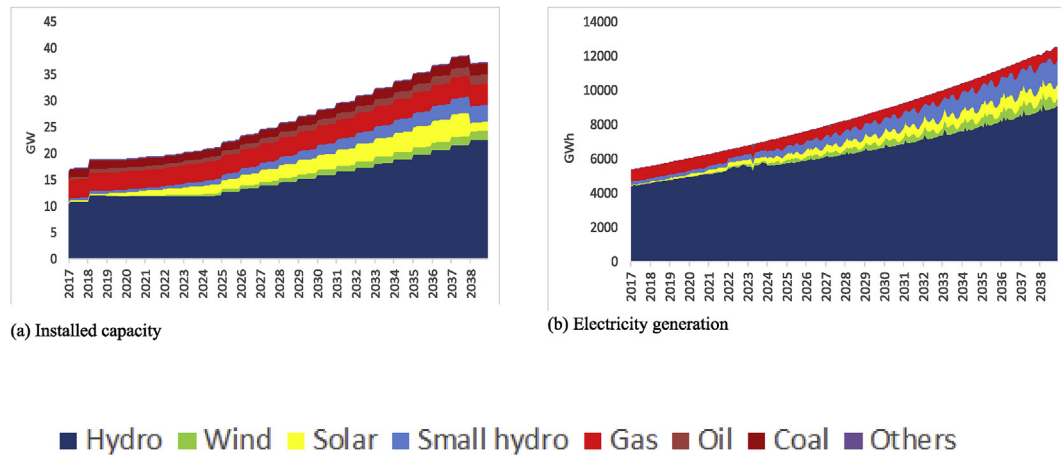


Fig. 11. (a) Installed capacity, with percentage of market share by technology to 2040 and (b) Electricity generation, in the RE&CM scenario.

prices will be.

This might make sense if complementarity between renewables is further accentuated by taking advantage of the possibility that hydro-power reservoirs might play the role of a backup to the system. What might be gained if the hydropower plant capacity increases slightly and other renewables increase by up to 30% of the total capacity. In this case, we also investigate likely gains in energy efficiency from the complementarity between renewables.

Fig. 12a shows the highest penetration of renewables, and Fig. 12b shows that most power generation comes from solar PV, wind, small hydro and hydropower from 2026, when thermal technologies are most available as a backup.

5.6. Scenario comparison

Simulation runs are presented separated into two groups – the more consistent and less volatile group, and the less consistent and more volatile group. Fig. 13 shows policy results for the BAU, More Renewable, and RE&CM scenarios. Under the BAU scenario system, system reserve margin oscillated between 8 and 15 per cent for most of the simulation period because of the capacity mechanism. This leads to low variability in the electricity price (Fig. 13b) resulting from the dynamic interaction between installed capacity and electricity demand. Hydroelectric capacity increases because of its low levelised cost, and thermal-power capacity (gas and coal-based technologies) grows with the support of the reliability charge mechanism in place. The dynamics

of emissions are presented in Fig. 13c, showing increasing emissions as the dispatch of fossil-fuelled technologies increases, conflicting with the goal of a 20 per cent CO₂ emissions reduction in Colombia (Ministerio de Medio Ambiente, 2015). Specifically, the contribution from natural gas technology increases leading to a significant increase in the total emissions of the system towards the end of the simulation.

Concerning the RE&CM scenario, the reserve margin oscillates due to an increase in renewables, and increases in the hydroelectricity capacity and thermal electricity (gas and coal-based technologies) investments through the capacity mechanism. Intermittency and seasonality induce price volatility. The dynamics of emissions are presented in Fig. 13b showing a reduction in emissions from 2023 as the contribution from renewable technologies increases. Specifically, increases in the contribution from wind, small hydro and solar PV technologies lead to a significant decrease in the system's total emissions towards the end of the simulation.

Under RE and free market scenarios, Fig. 13d shows thermal electricity (gas and coal-based technologies) and hydroelectricity all showing oscillations caused by scarcity prices when the system reserve margin is negative (Fig. 13a). Regarding CO₂ emissions, a free market exhibits the highest CO₂ emissions due to fuel oil and coal electricity generation (Fig. 13c), as thermal generation is necessary to supply the electricity demand during most of the simulation time. In the RE scenario, the dynamics of emissions are presented in Fig. 13c showing that emissions follow the behaviour of electricity prices, which indicates that the most expensive thermal generation is taking place. During most

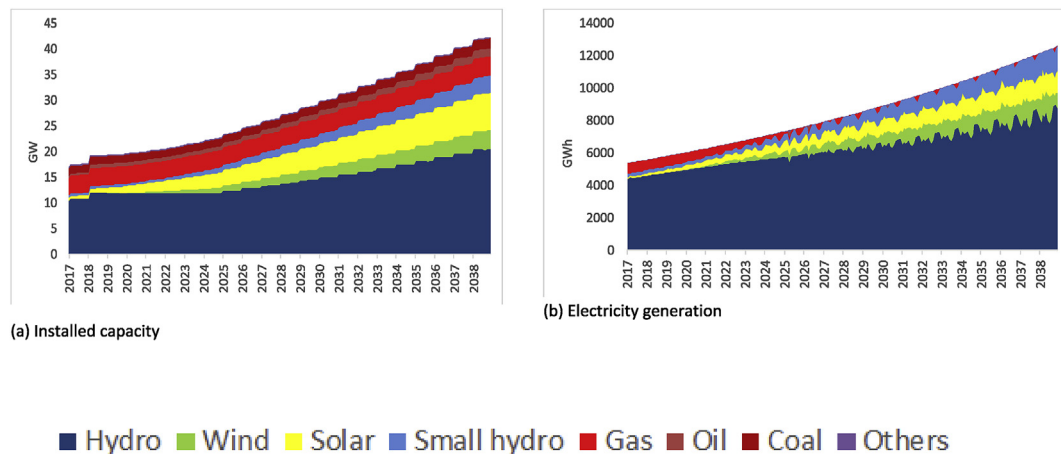


Fig. 12. (a) Installed capacity, with percentage of market share by technology to 2040 and (b) Electricity generation, in the More Favourable Renewables Component scenario.

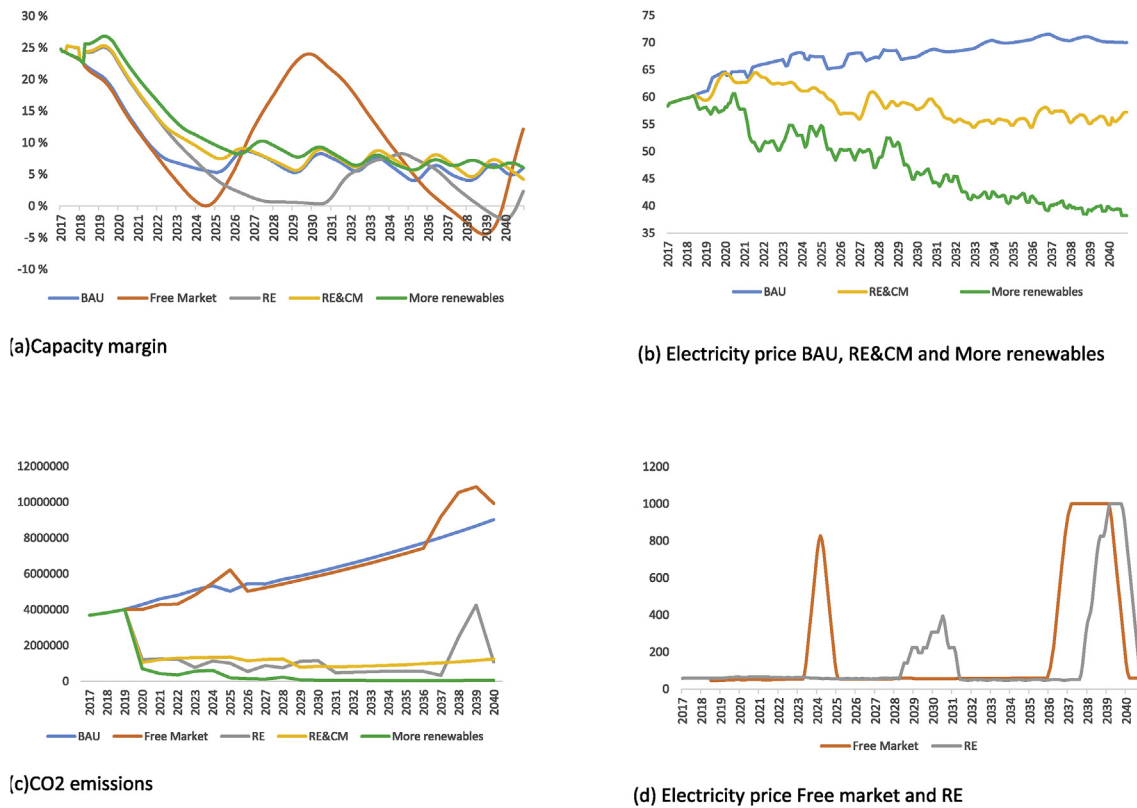


Fig. 13. (a) System reserve margin, (b) Electricity price BAU, RE&CM and More renewables (c) CO2 emissions and (d) Wholesale electricity price Free market and RE scenarios.

of the simulation, generation from renewable energy sources – wind, small hydro, and solar PV technologies – increases. Note, that while the electricity price is around 60 USD/MWh at the beginning of the simulation, it reaches almost 1000 USD/MWh by 2036.

Finally, the more favourable renewable scenario in Fig. 13 shows that: a) the system reserve margin oscillates between 5 and 15 per cent, much higher than that provided by policy in Scenario C (in which the system reserve margin even reached 0 per cent), which indicates electricity demand would most likely be met by higher water reservoir levels and rarely by thermoelectricity, which will be available for backup purposes; b) electricity price drops to about 43 USD/MWh given the little need for expensive thermal power, as shown in Fig. 13b; and c) CO2 emissions decrease greatly as renewable energy and hydro generation largely displace thermal generation. The More favourable renewable scenario has the best indicators performance (Fig. 13).

Fig. 14a and Fig. 14b show the complementarity between renewables – when solar and wind generation are at their highest level, water inflows are at their lowest, helping security of supply and reducing electricity prices (e.g. February 2017, 2018, 2019, 2037, 2038 and 2039). Furthermore, Fig. 14c and d show that during these months, while Sun and wind power are at their highest levels, reservoir levels drop continuously until April of the corresponding years. These figures show the complementarity between renewables and that this is more pronounced in the earlier years, when a higher percentage of renewables is in place. Note that during the period 2017–2019, the reservoir level drops to 63% while during the period 2037–2040 these levels remain above 70%. In these cases, sun and wind power help to preserve higher reservoir levels, allowing for more water inflows rather than using this source for hydroelectricity generation. This contributes to better reservoir management by saving water for long dry seasons.

Next, this paper establishes how solar and wind energy may contribute to a large hydro-based power system, such as the system in Colombia. Fig. 15 shows the evolution of the reservoir level (in

percentage terms) with and without the participation of solar and wind generation, for the scenario that is most favourable for renewables. As can be observed, when solar and wind are present, reservoir levels smoothly oscillate at high levels, but when they are not operational reservoir levels drop to lower levels.

Table 2 summarises the effects of the policies analysed here in terms of: total emissions, renewables market share, electricity price and system reserve margin. Emissions are measured as the total of emissions during the simulation time. The renewables market share represents the percentage of the installed capacity. The renewables generation share represents the percentage of renewable generation over the total electricity generation, at the end of the simulation period. Results indicate that a high renewables energy component performs better than other alternatives in all accounts. The significant participation of renewables and their complementarity contribute to both the lowest price as well as the lowest emissions, with the largest system reserve margin. Furthermore, as might be appreciated, this scenario makes better use of hydro power in reservoirs, keeping them at higher levels, by making lower use of them (72%). A small share of renewables helps by reducing price volatility but as this share becomes significantly larger, though the system reserve margin reaches high levels, it also brings much higher price volatility.

6. Conclusions and policy implications

This paper provides insights into the potential of renewable energy sources in the Colombian electricity supply industry. It examines the likely benefits of promoting renewable energies in the energy matrix and discusses the possibility of blackouts if the right policies are not in place.

Under the BAU scenario, the expansion of the electricity industry is primarily promoted by the capacity mechanism currently in place. This provides an objective system reserve margin, which explains increases

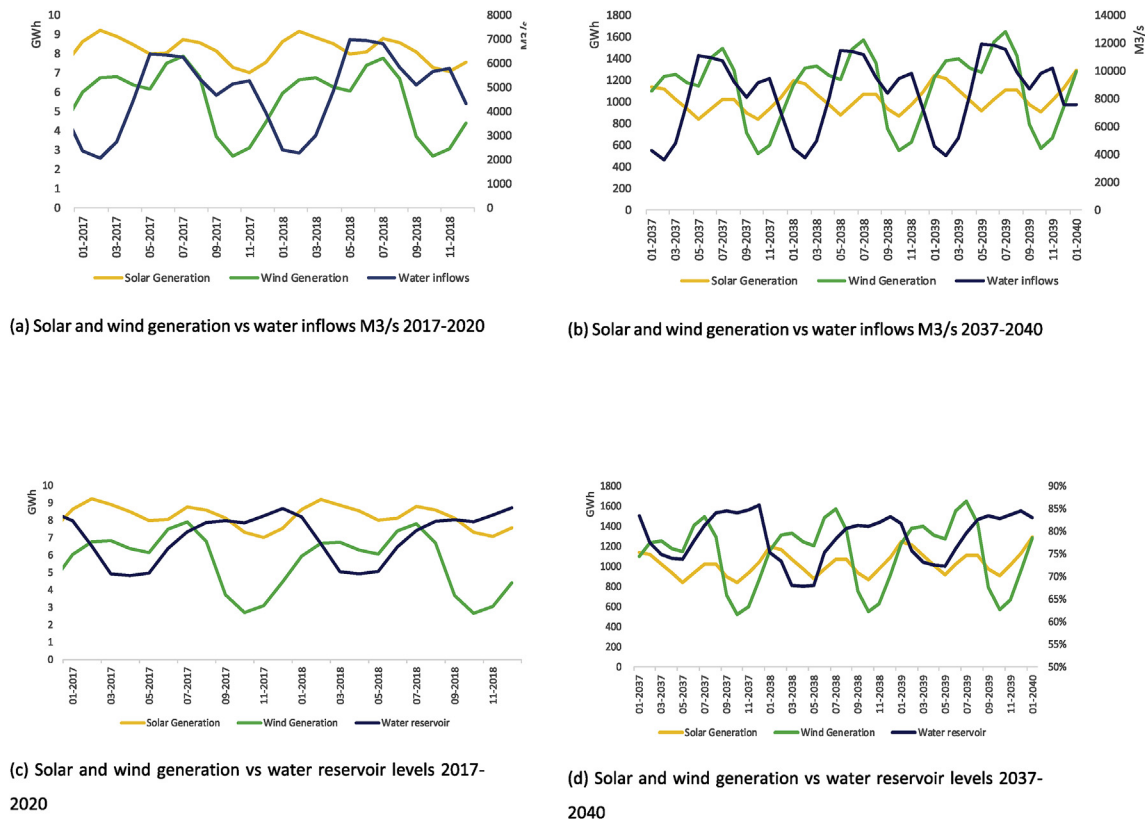


Fig. 14. (a), (b) Solar and wind generation vs Water inflows; (c), (d) Solar and wind generation vs Water reservoir levels.

in both pool prices as well as in CO₂ emissions. In this case, renewable energies are not significantly diffused in Colombia, despite their great potential, especially in wind and solar power. The removal of the capacity mechanism is unwise as it would result in a system with black-outs and induce high levels of emissions, high prices and high volatility in the system. Although the promotion of renewables reduces the levels of both blackouts and CO₂ emissions, this policy is unsustainable. Under these conditions, Colombia is better off with a policy on renewables than a completely free market, particularly because of the large hydroelectricity component in place.

Investigating further the role of renewables in energy systems, our study shows that solar, wind and hydro complementarity help security of supply at lower cost than thermoelectricity plants, and that in the mid-term the system becomes much cleaner and more balanced. Solar, hydro and wind complementarity leads to increases in energy efficiency

and may provide system reliability at a lower system reserve margin, saving water to deal with seasonality effects and very dry seasons (under El Niño conditions).

Under Scenario C, the system attains an unbalanced solution as it does not take sufficient advantage of the efficiency gains offered by the complementarity between renewables. This is attained in the scenario with a higher component of renewables. This paper concludes that when policy is carefully set, a high renewable energies scenario may be a good option for the mid-term as the complementarity between hydro, solar and wind lead to a high reserve margin, low price and very low CO₂ emissions, though at the expense of high price volatility.

Although counterintuitive, renewable energies help to increase security of supply in the Colombian electricity market because of their complementarity with hydropower technology. As incentives to invest in renewables achieve substantial diffusion of these technologies,

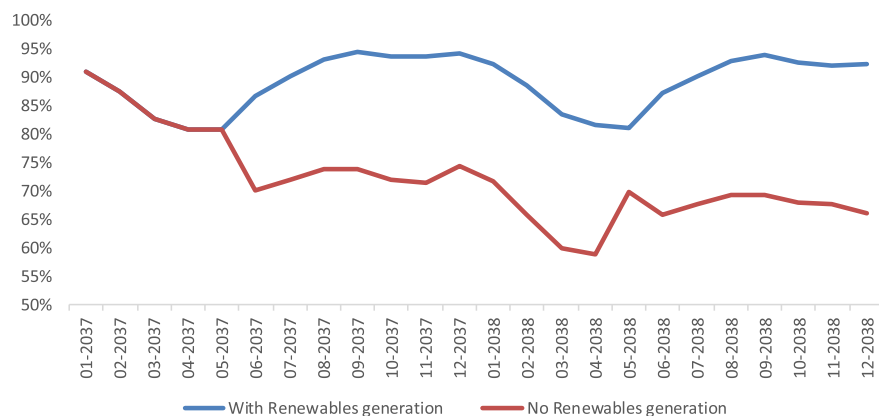


Fig. 15. Water reservoir levels comparison with renewables generation and no renewables generation.

Table 2
Scenarios summary.

Policy	Total Mtoe CO2 Emissions	Average wholesale electricity price USD/MWh	Min system reserve margin	Indicative standard deviation pool price USD/MWh	Renewables market-share	Renewables generation share	Hydro generation Share
BAU	145	65	7%	3	8%	7%	77%
No policies	159	205	–10%	4	13%	8%	74%
Renewable energies	157	151	–5%	5	22%	15%	72%
Renewable energies + BAU	81	60	5%	3.5	21%	13%	74%
High renewable energies component	44	55	11%	6.6	25%	19%	72%

electricity prices drop significantly, and thermal generation becomes less significant. This will be the case as the renewable energy potential in Colombia is high, and as these technologies exhibit lower costs with respect to conventional technologies.

The model is not exempt from limitations. First, uncompetitive behaviour between utilities is not considered, and pool price reflects an ideal power dispatch but, in any case, these effects should be dampened by the penetration of renewables. Second, neither long-dry nor flood seasons are taken into account in the model, nor changes in wind, solar and water inflow patterns. Furthermore, imports and exports of electricity are not incorporated.

The paper succeeds in its purpose on at least two counts: a) it shows the likely benefits and effects of renewable and traditional energies on the system, i.e. reduction in CO2 emissions and electricity prices; and b) it establishes the conditions under which renewables might increase system reliability. Further, a more in-depth analysis would provide avenues of research on how renewable energies in conjunction with hydro may provide system reliability, thanks to their complementarity.

Acknowledgment

The authors are grateful to EPM and COLCIENCIAS for providing the necessary financial support for this paper. The authors are also grateful to Paul Ellis and James Bleach, the obex project, for editorial work.

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